



ASPEN: Assisted-as-needed sEMG-triggered control of an upper limb exoskeleton for Post-stroke home rEhabilitationN

Development, verification and validation

Collaborative Robotics – Project Report

Advisors:

Prof. Ambrosini Emilia

Prof. Gandolla Marta

Tutor:

Dr. Pozzi Luca

Authors:

Bonini Nicolò

Di Guardo Roberto

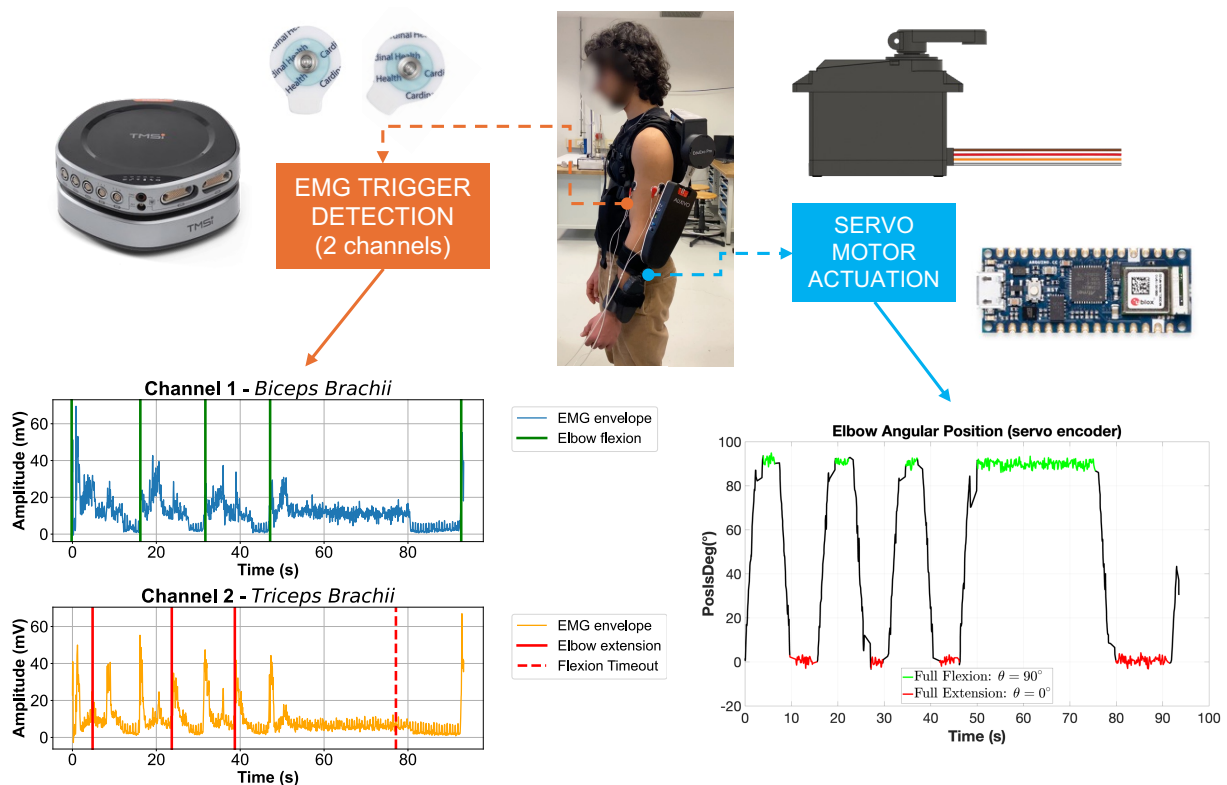
Scalia Nicolas

Sgorbati Simone

Speranza Matteo

Abstract and Graphical Abstract

ASPEN aspires to be an innovative, **“assisted-as-needed” system** designed for post-stroke rehabilitation of patients with mild upper-limb motor impairments. Combining Auxivo EduExo Pro exoskeleton with TMSi SAGA device, ASPEN utilizes surface electromyography (sEMG) signals to detect muscle activation, through a **binary controller algorithm with two channels (biceps and triceps)**. Those triggers are translated into **motion of the patient’s elbow (1 DOF)**, through the exoskeleton servo motor, **along predefined trajectories and at a customised speed**. The system offers three operational modes: “Empower”, which provides real-time assistance triggered by the patient’s muscle activation, “Stream”, enabling repeatable passive exercises via pre-recorded EMG data, and “Therapist”, for manual guidance during therapy sessions. These modalities address diverse rehabilitation scenarios, fostering active involvement and independent use. **Safety features**, such as a physical emergency stop button and a timeout for prolonged flexion, ensure the patient’s comfort and well-being. A simple and intuitive **Graphical User Interface (GUI) ensures ease of use**. The double-threshold algorithm for intention-detection achieved a **truthful detection of movement intentions** (>90% accuracy). Preliminary tests demonstrated the system’s ability to achieve higher accuracy in reaching the target angular positions (e.g., full flexion, 90°) and reduced variability compared to therapist-assisted movements (exo: RMSE = 4.53°, IQR = 0.93°; therapist: RMSE = 12.94°, IQR = 6.13°). This is a significant result since **high repeatability** of movements is known to be crucial to promote neuroplasticity, enhance motor learning, prevent incorrect compensations, monitor progress, and ensure a safe and effective recovery. Furthermore, usability tests on a group of five participants confirmed **ease of setup** (< 10 min.) and **portability** (median SUS score of 62.5/100).



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1. Motivation of the project

Stroke, or cerebrovascular accident, affects 15 million people worldwide each year, with 5 million individuals facing permanent disability [1]. The consequences of stroke depend on the location and extent of brain damage, which impacts motor abilities (e.g., hemiplegia) and neurological functions (e.g., aphasia). Upper-limb motor impairments, such as muscle weakness, joint laxity, changes in muscle tone, and reduced motor control, are particularly debilitating. These impairments restrict essential movements like reaching, grasping, and holding objects, making it difficult to achieve everyday tasks and significantly reducing independence in performing Activities of Daily Living (ADL) and the capability to perform social and occupational activities [2] [3]. A reduced arm dexterity is, ultimately, often associated to anxiety, depression a poorer perception of quality of life.

Rehabilitation is, therefore, vital for stroke patients, as it leverages neuroplasticity – the brain's ability to reorganize and adapt – to recover lost functions. While best practices emphasize early intervention, continuous monitoring and a multidisciplinary approach [4], traditional rehabilitation methods often lack sufficient engagement and personalization. These approaches can be monotonous and are not always adaptable to the specific needs of the patient, especially in home rehabilitation settings, where patients are expected to work independently.

2. Objective of the work

This project seeks to address these challenges by developing a rehabilitation device tailored for post-stroke patients with mild upper-limb disabilities (Muscle Research Council (MRC) score ≥ 2 for *biceps brachii* and *triceps brachii*). Using the Auxivo EduExo Pro, an educational exoskeleton kit, the system employs sEMG signals acquired through the TMSi SAGA device to trigger support. Integrating an "assist-as-needed" strategy, the system adapts to the patient's muscle activation, guiding them through predefined trajectories. By enhancing engagement and providing tailored assistance, the ASPEN system aims to improve rehabilitation outcomes, support functional recovery, and ultimately enhance the patient's quality of life [5].

Some further assumptions regarding the target patient group have been made:

- i. Patients can initiate voluntary movements of the fingers, hand and wrist (i.e., wrist flexors and extensors, fingers flexors and extensors, intrinsic hand muscles) to some degree, allowing the device to detect muscle activity for triggering assistive actions.
- ii. Patients' cognitive and motor abilities are sufficient to understand and follow simple rehabilitation tasks, ensuring meaningful engagement with the system.

3. Requirements

In designing this device, several requirements were considered. The most relevant ones related to the control scheme are here outlined and categorized as functional (F) or non-functional (NF).

REQUIREMENT	TYPE	DESCRIPTION	MEANS OF VERIFICATION (MoV)
Intention Detection from EMG signals	F	The system must detect movement intentions from the relevant muscle groups EMG signals. Onset detection must achieve an accuracy of $\geq 90\%$ (i.e., correctly identifying $\geq 90\%$ of triggers/de-triggers).	Simulate EMG for at least 20 repetitions varying input type (e.g., isometric / dynamic contraction, unilateral / bilateral, quick / slow) and measure onset detection accuracy ($\geq 90\%$ correct identifications).
Target Point Accuracy	F	The system must reach target angular positions for the elbow joint with: - an accuracy: $RMSE = \sqrt{\sum_{i=1}^n \frac{(\theta_{target} - \theta_{measured})^2}{n}} [^\circ]$ - a variability: $IQR = Q_{3\theta_{measured}} - Q_{1\theta_{measured}} [^\circ]$ not greater than that of a therapist guiding the patient movement ($RMSE_{exo} < RMSE_{therapist}, IQR_{exo} < IQR_{therapist}$)	Measure the position using the motor encoder for at least 20 repetitions and compute accuracy and variability measures with respect to the target position. Compare exoskeleton-assisted (experimental) and therapist-assisted (control) cases.
Safety Mechanisms	F	The system must include a physical emergency stop button, easily accessible and operable by the patient or therapist. Furthermore, in case of prolonged elbow flexion (> 30 seconds), a timeout should trigger automatic relaxation or cessation of motor actuation to prevent muscle strain or discomfort.	Test the emergency stop functionality in 3 simulated scenarios, ensuring immediate cessation of movement and servo detachment (target response time ≤ 1 second) in case of manual button pressing and prolonged elbow flexion (> 30 seconds). Verify proper motor shutdown and confirm no residual movement occurs in all scenarios. Verify timeout activation for the prolonged flexion condition.
Real-time Motor Actuation via EMG triggers	NF	The system must translate the detected movement intentions into motor actuation with a delay of < 0.500 s and an accuracy of $\geq 80\%$ (i.e., correctly translating $\geq 80\%$ of triggers/de-triggers into motion).	Simulate EMG for at least 20 repetitions, measuring detection-to-actuation latency (target: < 0.500 s) and motor actuation accuracy ($\geq 80\%$ correct translations).
Ease of Configuration	NF	Non-technical users (e.g., patients, caregivers) should be able to configure the system in under 10 minutes using an intuitive interface.	Conduct configuration trials with at least 3 non-expert users and document the average time required (target < 10 minutes).
Usability and Comfort	NF	The system (exoskeleton + electrodes) must be donnable / doffable within 5 minutes . The contact points must use hypoallergenic, breathable materials to ensure comfort during extended sessions. The overall experience must be satisfactory for the patient.	Conduct usability tests with at least 3 users (patients), ideally representing a diverse range of features, ages, genders, and impairments. Measure donning / doffing times, and gather comfort ratings via questionnaires, translating the results into a 100-point scale (with a target score of ≥ 60), based on the System Usability Scale (SUS).

4. Materials and methods

4.1 Hardware

ASPEN exploits three main hardware components, as depicted in *Figure 1*, used in synergy:

- **TMSi SAGA:** the device is used to acquire high-resolution EMG signals, detecting muscle activity. The device is connected to the processing system via a USB cable playing a critical role in interpreting the user's intention, by using its data to trigger the exoskeleton's support.
- **EduExo Pro Exoskeleton:** this wearable exoskeleton provides a single degree of freedom for elbow flexion and extension, replacing human hinge joint with a robotic rotational joint. Actuation is achieved through a servo motor controlled by an Arduino Nano 33 IoT, which is exploited to receive commands based on processed EMG signals. The EduExo Pro has an integrated IMU sensor on the upper arm, an encoder in the motor at the elbow joint and a load cell at the wrist. The hardware supports both Wi-Fi and USB connectivity.
- **PC:** it serves as the central hub for connecting TMSi SAGA and EduExo Pro. It is responsible for running, through Visual Studio Code, the Python scripts for signal processing within a virtual environment, and the Arduino IDE (v 2.3.3) for programming the exoskeleton's control logic.



Figure 1

4.2 Kinematic and dynamic model

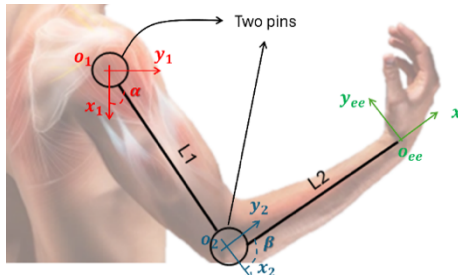


Figure 2

KINEMATIC MODEL. The shoulder and elbow are the key joints of the upper limb. In 3D space, the shoulder provides 3 degrees of freedom (DOFs), and the elbow provides 1 DOF. The upper limb can be approximated in 2D as two beams connected by two pins, forming a configuration as shown in *Figure 2*. The combined system of the arm and exoskeleton can be approximated as a SCARA mechanism,

given their similar structure and movement characteristics. The DOFs are calculated by considering the total DOFs of a beam structure in 2D space and subtracting the degrees of constraints (DOCs) imposed. In 2D space, the system has 6 DOFs (from the two beams) and 4 DOCs (from the two pins), resulting in 2 DOFs. However, since the shoulder joint is not motorized, its angle (α) can only be manually adjusted by the user. Therefore, only the elbow's angle (β) is considered for this model, resulting in 1 DOF. The exoskeleton includes mechanical constraints that limit its range of motion such that the maximum achievable β is approximately 100° , which is designed to match the natural human range of motion, approximately 130° .

In the exoskeleton control, forward kinematics was used to predefine the elbow joint's angular position. By setting the desired joint angle, the end-effector's position was determined using specific motion equations:

$$\begin{cases} X_{ee} = L_1 \cdot \cos(\alpha) + L_2 \cdot \cos(\alpha + \beta) \\ Y_{ee} = L_1 \cdot \sin(\alpha) + L_2 \cdot \sin(\alpha + \beta) \end{cases}$$

where X_{ee} and Y_{ee} are the coordinates of the end-effector (EE), located at the wrist, as presented in *Figure 2*, with respect to the first reference system ($x_1 \rightarrow$ vertical axis, $y_1 \rightarrow$ horizontal axis).

DYNAMIC MODEL. As shown in *Figure 3*, the forces acting on the system consist of the weights due to the mass of the exoskeleton $m_{exo} \cong 2$ kg and the arm, applied at the centre of mass of each link. Additionally, the system includes the load of the object held by the user (e.g. a bottle). The model can be scaled based on the subject's anthropometric data (e.g. reference weight $M = 70$ kg which includes also m_{exo} , and the estimated quantities shown in table, based on [6]).

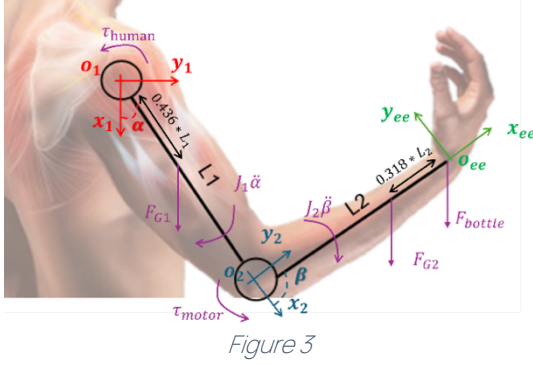


Figure 3

Quantity	Value	Units	Description
m_1	$0.028 \cdot M$	[kg]	mass of forearm
m_2	$0.022 \cdot M$	[kg]	mass of forearm and the hand
m_{bottle}	0.7	[kg]	mass of bottle to lift
g	9.81	[m/s ²]	gravity coefficient
l_{G1}	$0.436 \cdot L_1$	[cm]	barycentre starting from the shoulder
l_{G2}	$0.682 \cdot L_2$	[cm]	barycentre starting from the elbow
J_1	$m_1 \cdot L_1^2$	[kg·m ²]	inertia of upper arm
J_2	$m_2 \cdot L_2^2$	[kg·m ²]	inertia of forearm and the hand

The exoskeleton is attached to the upper body via a textile vest that houses a rigid frame. At the shoulder, an integrated torsion spring assists the user in lifting the arm.

The dynamic model was developed starting from the general Euler-Lagrange equation for an n-DOF system, which defines the forces and torques needed to produce motion:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + f(q, \dot{q}) + G(q) = \tau - J(q)h_{cart}$$

$M(q)$: inertia matrix accounts for the mass distribution, and it determines the resistance to acceleration.

$C(q, \dot{q})$: coupling between joint velocities, generating Coriolis and centrifugal forces due to motion.

$G(q)$: effect of gravity acting on the links.

$f(q, \dot{q})$: friction and viscous velocity (neglected).

τ : joint torque generated by the motor to win the other terms and satisfy angle in input.

$J(q)h_{cart}$: external input applied to the robot usually on EE by environment or due to contact with a user.

$$\begin{bmatrix} m_1 l_{G1}^2 + m_2 (L_1^2 + l_{G2}^2 + 2L_1 l_{G2} \cos(q_2)) + J_1 + J_2 & m_2 (l_{G2}^2 + L_1 l_{G2} \cos(q_2)) + J_2 \\ m_2 (l_{G2}^2 + L_1 l_{G2} \cos(q_2)) + J_2 & m_2 l_{G2}^2 + J_2 \end{bmatrix}$$

$$\begin{bmatrix} m_2 L_1 l_{G2} \dot{q}_2 \sin(q_2) & -m_2 (\dot{q}_1 + \dot{q}_2) L_1 l_{G2} \sin(q_2) \\ m_1 L_1 l_{G2} \dot{q}_1 \sin(q_2) & 0 \end{bmatrix}$$

$$\begin{bmatrix} (m_1 g l_{G1} + m_2 g L_1) \cos(q_1) + m_2 g l_{G2} \cos(q_1 + q_2) \\ m_2 g l_{G2} \cos(q_1 + q_2) \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\tau_{exo} \text{ [N·m]}$$

$$\begin{bmatrix} 0 \\ -F_{bottle} \cdot L_2 \end{bmatrix}$$

The kinematic and dynamic models presented here have the potential to support the implementation of a compliant and transparent control of the exoskeleton. However, due to observed limitations in the reliability of the force sensor during preliminary testing, their integration into the current project is not yet feasible.

4.3 Control design

Hardware level of compliance

As a collaborative-wearable robot, the exoskeleton shell, which is physically attached to the user, serves as the interaction point between the torque generated by the motor (τ_{exo}) and the torque produced by the user's muscles (τ_{muscle}). During the initial phase of rehabilitation, the ASPEN device is designed to guide the movement entirely on its own, with no

Anatomical Joint	Joint type	Actuation Hardware	Sensing Hardware
Shoulder	Soft	Passive: Torsion spring	No sensors
Elbow	Stiff	Powered: Electrical motor	Joint space: Encoder

contribution from the user ($\tau_{muscle} = 0$), even if the hardware is built to be compliant and responsive to the input.

Indeed, the shoulder, implemented as a soft joint, simplifies the control strategy with mechanical compliance, without the inclusion of additional sensors. In contrast, the elbow, being a rigid actuated joint, requires a suitable sensor network to detect voluntary movement. The motor encoder is employed as the sensing hardware in joint space, while the integrated force sensor located at the end-effector in task space is excluded, due to observed limitations in reliability.

Low-level control solution

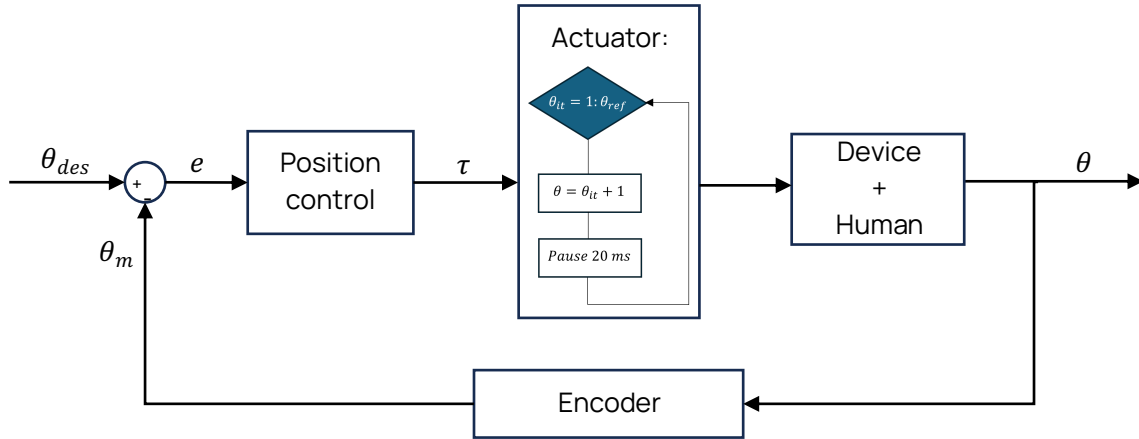


Figure 4

In ASPEN system, a low-level position control scheme for the state θ [°] is implemented. The control scheme illustrated in *Figure 4*, relies on defining a desired position (θ_{des}) set by the user once for the whole rehabilitation session. This desired position is calculated as $\theta_{des} = \theta_{target} + \theta_{offset}$, where θ_{target} is the target position in the anatomical system, θ_{offset} accounts for the orientation of the servo motor relative to the anatomical system and represents the deviation from the anatomical reference frame. θ_{des} is the position that the servomotor reads and actuates, and it is compared in the control with the position measured by the sensor (θ_m), in this case an encoder. At each iteration a position error is computed as $e = \theta_{des} - \theta_m$. If the position error falls within a specified tolerance ($e < 10^\circ$), the motor is detached, otherwise this error serves as the input for the position control.

The position control provides the necessary torque (τ) to reach the correct configuration through an inner loop (“Actuator” block) in which at every iteration the angle is increased by 1° . The iteration delay (*Pause*) is set based on the desired angular velocity, which is computed as: $\omega_{robot} = \frac{1^\circ}{time} = \frac{\pi}{180 \cdot time} [\text{rad/s}]$. Specifically, for fast motion, a 10 ms interval generates a velocity of 1.74 rad/s, while a 20 ms interval is used for slower motion, resulting in a velocity of 0.873 rad/s. Both values align with the typical range of human joint motion, which occurs at approximately at 1÷2 rad/s [7]. This custom control approach relies on the position control mechanism already integrated into the servo motor [8], although the details of its operation are not accessible, with the custom implementation providing redundant functionality.

Additionally, a physical button placed on the upper arm cover (*B2*, in black), when pressed, immediately terminates the Wi-Fi connection with the movement trigger source (as described in

the next paragraph). This action halts any ongoing movement and detaches the servo, providing a physical emergency safety mechanism.

High-level control solution

The high-level control strategy is related to the desired human and robot interaction during exercise. A Graphical User Interface (GUI) has been developed to assist the therapist in various stages of rehabilitation. The GUI is divided into two main sections. The first section focuses on system setup, featuring a "Record" button for saving EMG signals and a "Calibrate" button for evaluating muscle activation thresholds (see "Intention detection strategy" section). The second section includes a "Stop" button that terminates any active task and returns the exoskeleton to its resting position. Finally, each modality has its own button:

- "Therapist" mode, in which the exoskeleton is backdrivable, the motor is detached, avoiding resistance to external inputs. Since it uses an electrical motor, there may be a residual torque to be added to the natural inertia, resulting in an initial force that needs to be overcome for intentional movements. This configuration allows the therapist to move the patient's arm freely. Additionally, the robot can still receive manual triggers (i.e., inputs for motor actuation) through a GUI, for both movements (elbow flexion / extension).
- "Stream" is a passive-triggered modality, where the triggers are derived from a previously recorded biceps EMG signal. The ASPEN device comes with a set of prerecorded movements but can be customised and enriched with movements recorded in the training sessions. This allows for high repeatability of the training exercise even without the TMSi SAGA device.
- "Empower" is a passive-triggered [9] mode, where the assistance is triggered by real-time EMG measurements coming from the TMSi device. This mode does not require effort from the user except for muscle activation for intention detection. Once the system detects the user's intention to move their arm, the robot guides them along a predefined trajectory.

The use of passive-assistance modalities helps prevent secondary complications such as stiffness in soft tissues and muscle spasticity. Passive mobilization stimulates somatosensory input, potentially enhancing brain plasticity and improving the patient's musculoskeletal condition. However, passive modalities also have significant limitations. A primary concern is that without active involvement from the patient, long-term recovery may be compromised.

Intention detection strategy

The system uses EMG signals to detect the user's intention during "Empower" mode. Due to a latency of approximately 0.150 s between the onset of muscle activation and the subsequent movement (electromechanical delay), the user's intention can be detected during this time window, allowing the exoskeleton to be actuated accordingly.

In this system, two bipolar channels are used, requiring five electrodes to be placed on the user, upon proper skin preparation with rubbing alcohol, following the SENIAM guidelines [10]. Two electrodes are positioned on the left biceps to detect the user's intention to flex the elbow, while two other electrodes are placed on the left triceps to detect the user's intention to extend the elbow. A reference electrode is positioned on the bony structures of the elbow, specifically

either the medial epicondyle of the humerus or the olecranon. The medial epicondyle is generally preferred due to the absence of loose tissue and movable skin during flexion and extension. The triceps is the ideal choice because it directly controls elbow extension, making it the most natural and efficient muscle for this movement. However, the system has also shown good results with alternative muscles, such as the wrist flexors and extensors, although their use is less natural since they are not directly involved in elbow extension, resulting in a less intuitive control. The quality of the signal measured with surface electrodes is affected by several factors, including the positioning of the electrodes relative to the muscle and skin impedance.

An alternative approach for detecting elbow extension intent involved using the wrist force sensor, which would have improved compliance but was not implemented due to its unreliability.

The initial phase of the rehabilitation session with ASPEN involves recording the patient's muscular activity ("Record"), with a sampling frequency of 500 Hz. In the calibration phase ("Calibrate"), the EMG signal is processed as a stochastic signal. The EMG envelope from raw EMG measured by TMSi SAGA device (in Poly5 format) is derived via pre-processing Python functions, *ad hoc* developed. The raw signal is filtered using a fifth-order Butterworth bandpass filter, with a low cut-off frequency of 10 Hz and a high cut-off frequency of 100 Hz. The rectified EMG signal is obtained by taking the absolute values of the EMG raw signal. This rectified EMG signal is filtered using low frequency fifth-order Butterworth filter with a cut-off at 5 Hz.

A binary controller is implemented; therefore, thresholds are required to discriminate between intentions (towards elbow flexion versus extension). Specifically, a double threshold algorithm is chosen. The first threshold type, $th1$ [mV], is based on the EMG envelope amplitude and is computed for each channel; above this threshold muscle activation is detected. In particular, from the resulting EMG envelope of each channel, at least three baseline segments (i.e., portions of the signal in which the muscle is at rest) are considered. From each segment, a value is calculated by adding three times the standard deviation to the mean value of the samples. These values are used to determine three potential values for $th1$: the maximum value, the minimum and the median of the three. In this work, the maximum is chosen (see "Onset Detection Accuracy" test). The second threshold, common to both channels, $th2$, set fixed at 0.200 s, is used to validate the intention before triggering the exoskeleton movement.

It is important to note that each time a different tester uses the exoskeleton, $th1$ must be recalculated, while $th2$ remains constant. Additionally, every 30 minutes, a pop-up message prompts the user to recalibrate the EMG repeating this procedure, as the signal can fluctuate significantly due to factors such as electrode movement, dryness, and muscle fatigue.

During the "Empower" mode, raw EMG data are continuously analysed and filtered in real-time by a Python code, using the same pre-processing functions developed for EMG calibration. Once the double threshold conditions are met, a trigger is sent to the Arduino board, leading to the actuation of the exoskeleton. Additionally, a safety mechanism is implemented. Once the elbow has completed flexion, if no triceps activation is detected within 30 seconds, the system

automatically returns the arm to a resting position (elbow extension) to avoid prolonged muscle tension or joint stress. This timeout duration can be customised based on user preferences.

Since “Stream” mode aims to emulate a real-time application, the pre-recorded raw EMG signal from a single channel (biceps) is analysed with the same pre-processing functions discussed above. Flexion movement intention is detected using the same double threshold method as in “Empower” mode (i.e., trigger if amplitude is above $th1$ mV for $th2$ seconds). Instead, extension movement is triggered when the amplitude of the signal is below $th1$ mV for $5 \times th2$ seconds.

4.4 Protocol for Testing and Validation

This section describes the experimental protocols and measures used to assess components and the fully integrated ASPEN system, ensuring adherence to requirements and functionality.

Component Test Protocol - Onset Detection Accuracy

This test validated the EMG pre-processing algorithm, focusing on a single channel, and aiming for an onset detection accuracy of at least 90% to meet the requirement. Five EMG signals were analysed, and the three different potential values (max, min, median) for the threshold $th1$ were compared, to identify the most effective approach in accurately detecting muscle activations.

Component Test Protocol - Target Point Accuracy

This test evaluated the exoskeleton's ability to guide the elbow joint to target angular positions with greater accuracy (RMSE) and lower variability (IQR) compared to therapist-assisted movements, exploiting the backdrivability of the exoskeleton, to meet the requirement. Using encoder data, the RMSE and IQR were computed and compared across two different subjects (subject 1 and subject 2), two different ROMs (reduced 0° - 45° and extended 0° - 90°) and two different motion speeds (slow at 0.873 rad/s and fast at 1.74 rad/s), for three repetitions each.

System Test Protocol - Delay and Motor Actuation Accuracy

This test evaluated the delay between the system detecting EMG activation and the exoskeleton initiating movement. To meet the requirement, the response latency had to remain below 0.500 s. Additionally, the motor actuation accuracy – defined as the percentage of correct translations of EMG triggers into movements – was assessed, with the target being a success rate of at least 80%. Five different measurements were used.

System Test Protocol - Ease of Configuration Evaluation

This test focused on measuring and comparing the setup time required for five non-expert users. To satisfy the requirement, the median setup time had to remain below 10 minutes.

System Test Protocol - Usability and Comfort Assessment

This test assessed the system's usability and comfort through a survey (available [here](#)) conducted with five participants. The survey included demographic questions, ten closed-

ended SUS questions, and two open-ended questions for feedback. A SUS score was calculated for each participant based on their responses to the closed-ended questions. To meet the usability requirement, the median SUS score needed to exceed 60.

5. Results

This section presents the results of the five tests conducted to validate ASPEN requirements, as outlined in the previous protocol section. Any deviations from the requirements are analysed, and potential future improvements are discussed.

Component Test Results - Onset Detection Accuracy

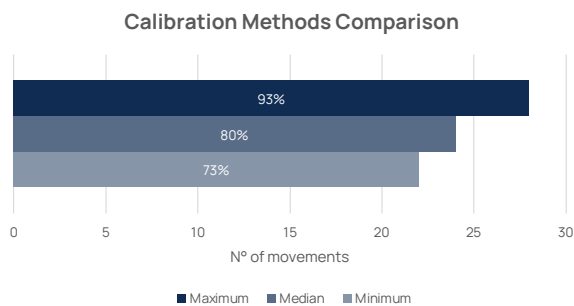


Figure 5

In this test, to choose the best value for the threshold $th1$ (referred to the double threshold algorithm discussed in “Intention detection strategy”) among the three options (max, min, median), five EMG signals were run offline for each alternative, and the accuracy of movement intention detection was evaluated by the ratio of correctly detected movements to the total.

Considering 30 slow isometric movements across the five signals, the maximum method proved to be the most accurate in correctly detecting muscle intentions and was the only one to achieve a success rate above 90%, as indicated in Figure 5. This is because, with a higher $th1$ value, it is more conservative, making it less susceptible to noise in the signal which could cause the algorithm to misinterpret the intention of the user. However, despite being more conservative, it remains prone to errors in those cases where the signal experiences significant fluctuations, but still outperforming the other two methods. Consequently, the maximum $th1$ was selected as the calibration method in the system, since it was the only one that fulfilled the requirement.

Component Test Results - Target Point Accuracy

This test compared the exoskeleton’s movement precision and variability against the therapist-assisted performance across different scenarios. Each measurement was repeated three times, leading to a total of 24 measurements. For each repetition, the user was asked to perform three complete flexions and extensions. Each measurement was saved in a CSV format file (i.e., 24 CSV files) and analysed using a MATLAB algorithm. For each measurement, the data were first filtered

using a low-pass filter (fourth-order Butterworth) at 3 Hz. Then, the algorithm identified the plateaus (i.e., when the movement reached its target point).

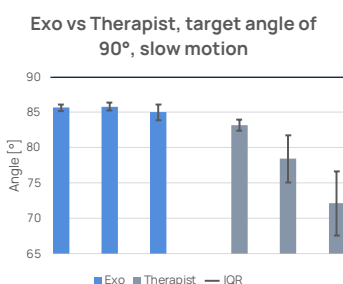


Figure 6

From the plateaus, the angular positions in degrees were extracted, and the median and IQR values were computed separately for the rest (0°) and contracted positions (90°/45°). Thus, for each file, two sets of metrics (median and IQR) were obtained — one for the rest position

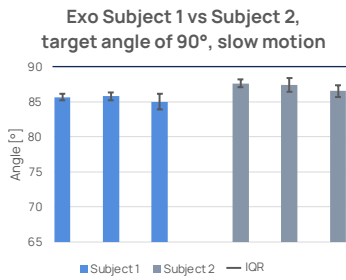


Figure 7

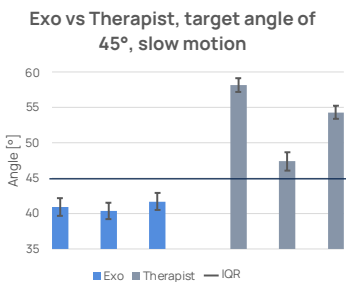


Figure 8

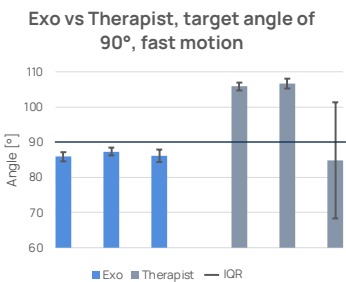


Figure 9

and one for the contracted position. Additionally, the RMSE was calculated using the median values across all files related to a specific measurement to evaluate the accuracy in reaching the target positions.

The first test compared the movements executed by the same subject, in reaching a target flexion angle of 90° in two cases: guided by the exoskeleton (experimental) versus therapist-assisted (control). As shown in Figure 6, the exoskeleton-guided case, achieved a lower RMSE (4.53°) compared to the therapist-assisted case (12.94°). The same applied for the IQR, which was consistently lower for the exoskeleton (average of 0.93° vs therapist's average of 6.13°). Figure 7 shows a comparison between two different subjects, yielding similar results and confirming the validity of this test.

In a second scenario, the ROM was reduced, setting the final angle to 45° . Once again, as evidenced by Figure 8, the RMSE of the exoskeleton-guided (4.05°) was lower compared to the therapist-assisted (9.41°), indicating lower variability. These findings were also observed in a last comparison, where the angular velocity was increased, both with robot and therapist (Figure 9). Additionally, these same analysis, when applied to the resting angle (0°), instead of the target angles ($90^\circ/45^\circ$), showed similar results.

In conclusion, the requirements were met in this test. This is since the exoskeleton's motion is robustly controlled by the system's algorithms, which ensure consistent behaviour and reduce variability. In contrast, therapist-assisted movements are variable because the "algorithms" of human motion control rely on individual experience and motor memory which affect the capability to regulate motion parameters (speed, articular angles, torques).

System Test Results - Delay and Motor Actuation Accuracy

This test evaluated the robot's movement initiation and delay performance using a Wi-Fi connection. Trigger timestamps were recorded in a CSV file, and sensor data from the encoder were analysed with a MATLAB algorithm to calculate the delay by comparing trigger times to movement initiation times. Data from 5 files (20 movements total) showed that the robot

successfully initiated movement in 80% of cases, meeting the motor actuation requirement.

However, as shown in Figure 10, the median delay in reaching the target position (90°) across all files was 1.114 s, exceeding the required 0.500 s latency, with a similar result (1.065 s) for reaching the rest position (0°).

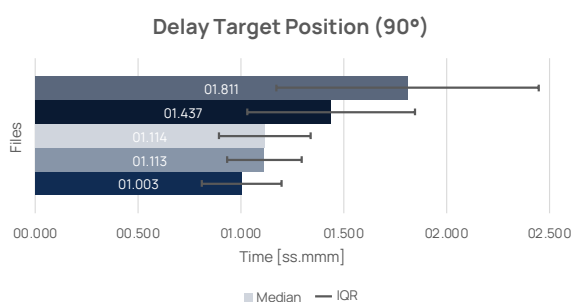


Figure 10

To improve performance, future tests could explore a wired connection to reduce communication variability. While falling short of the requirement, this result is a reasonable starting point for further refinement.

System Test Results - Ease of Configuration Evaluation

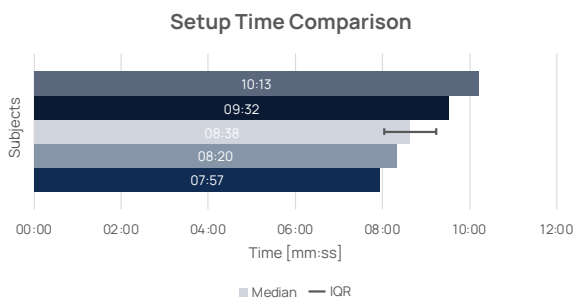


Figure 11

This test measured the time required to setup the system for five subjects. As illustrated in *Figure 11*, the median setup time was 08:38 minutes, with an IQR of 01:02 minutes, meeting the requirement of being under 10 minutes. Testing the setup process with a larger and more diverse group of users, including different ages and features (i.e., digital literacy, muscle impairment), could provide further insights.

System Test Results - Usability and Comfort Assessment

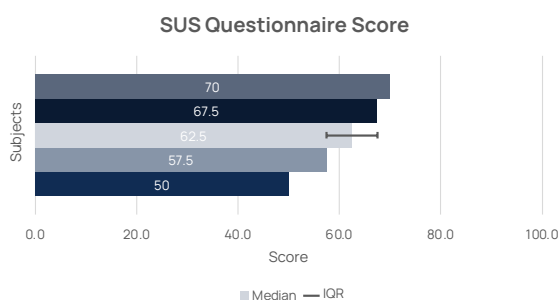


Figure 12

This test assessed the usability and the comfort of the system. All five participants were male, aged 18–30 years, with heights ranging from 171 cm to over 190 cm. Additionally, all subjects had a MRC score of 5, indicating normal muscle strength. As shown in *Figure 12*, the median SUS-questionnaire score was 62.5, with an IQR of 10, meeting the requirement of exceeding 60. Positive open-ended feedback highlighted the system's ease of use, while the main negative aspect was that the robot's movements appeared jerky at times.

Once again, this test should be repeated with a larger, more diverse sample, including participants of different genders, ages, and particularly varying MRC scale capabilities.

6. Discussion and Conclusion

ASPEN aimed to develop an exoskeleton-based rehabilitation system for post-stroke patients. It successfully achieved key objectives, including real-time intention detection (93% success rate) and improved target angular positions reaching compared to therapist-assisted movements (e.g., in the 90°/slow motion case, for three measurements, RMSE = 4.53° for the exoskeleton-guided vs RMSE = 12.94° for the therapist-assisted).

Despite overall progress, several challenges affected system's performance and development. Although the motor actuation accuracy successfully met the target of 80%, a median delay of 1.114 s persists between detection of the intention of movement and actuation. Testing a wired connection may help mitigate this delay, though it would restrict system's mobility. Additionally,

hardware reliability issues, such as the inconsistent performance of the force sensor, have hindered the development of a compliant and transparent control strategy for the moment.

Looking forward, several creative avenues for development are envisioned:

- i. **Advanced Modes:** Introducing “Progress” (active) mode to encourage patient effort with proportional-EMG support, and “Challenge” (resistive) mode for strength training in advanced rehabilitation phases. To support future rehabilitation phases with active patients, it would be essential to leverage the hardware's compliance, by exploiting the kinematic and dynamic models previously presented.
- ii. **Adaptive Learning:** Implementing machine learning algorithms to dynamically calibrate the system's thresholds *th1* based on the real-time EMG signal, to improve accuracy of muscle activation identification.
- iii. **Longitudinal Studies:** Conducting long-term clinical trials to evaluate the system's efficacy in improving motor recovery and quality of life over extended rehabilitation periods, while also testing the system's requirements with a more diverse pool of participants.

In conclusion, ASPEN passive modalities enable precise, repetitive movements, fostering neuroplasticity and reducing the physical burden on therapists. By refining the current system and exploring the proposed developments, ASPEN has the potential evolve into a comprehensive and effective solution for "assisted-as-needed" rehabilitation.

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